

```
library(tidyverse)
library(plotly)
library(ggplot2)
library(ggfortify)
library(ggpubr)
library(sciplot)
library(tigr)
library(knitr)
```

Abstract

This survey analysis was performed with scales obtained from the Beck's Cognitive Insight Scale (Beck, Bauruch, Baller, Steer, & Warman, 2004) and The Centrality of Religiosity Scale (Huber & Huber, 2012) to see if there is a relationship between the participants Insight and Religiosity. This survey collected 45 participants. Additionally, the study looked at the relationship between Age and Gender with Insight and Religiosity scores. In addition to a plot that shows the approximate scores of each participant a linear regression was performed. In addition the regression analysis a One-Way ANOVA were performed to examine the similar concept. A One-Way ANOVA were then Performed followed by an ANCOVA and MANOVA. There were no significance in any of the analysis performed. This could be due to the small sample size and poor diversity among the population.

Methods

A bar graph were used to represent the overall scores of each participant. A regression analysis were then performed to analyze the relationship between the Religiosity (R Totals) and Insight (I Totals) scores. A One-Way ANOVA was then performed to also assess the relationship between Religiosity and Insight Scores. This was then followed by a Three-Way ANOVA to assess the relationship between the Religiosity scores, Age, and Gender to the Insight scores. Lastly, A MANOVA analysis were performed to combine the Religiosity and Insight scores to compare the relationship with the Age and Gender variables.

Results

summary (FaithandInsight)									
		Gender		Age		I1			
Subnum						Min.			
Length:45		Min.	:1.000	Min.	:1.000	Min.	:1.000		
Class :character		1st Qu.:	:2.000	1st Qu.:	:1.000	1st Qu.:	:2.000		
Mode :character		Median :	:2.000	Median :	:1.000	Median :	:2.000		
		Mean :	:1.778	Mean :	:2.267	Mean :	:2.356		
		3rd Qu.:	:3.000	3rd Qu.:	:4.000	3rd Qu.:	:3.000		
		Max. :	:3.000	Max. :	:5.000	Max. :	:4.000		
		I2		I3		I4		I5	
Min.	:1.000	Min.	:1.000	Min.	:1.000	Min.	:1.0	Min.	:1.000
1st Qu.:	:2.000	1st Qu.:	:1.000	1st Qu.:	:2.000	1st Qu.:	:1.0	1st Qu.:	:2.000
Median :	:2.000	Median :	:2.000	Median :	:2.000	Median :	:2.0	Median :	:2.000
Mean :	:2.333	Mean :	:1.711	Mean :	:2.356	Mean :	:1.8	Mean :	:2.244
3rd Qu.:	:3.000	3rd Qu.:	:2.000	3rd Qu.:	:3.000	3rd Qu.:	:2.0	3rd Qu.:	:3.000
Max. :	:4.000	Max. :	:4.000	Max. :	:4.000	Max. :	:3.0	Max. :	:4.000
		I7		I8		I9		I18	
Min.	:1.000	Min.	:2.000	Min.	:1.000	Min.	:1.000	Min.	:1.000
1st Qu.:	:1.000	1st Qu.:	:2.000	1st Qu.:	:2.000	1st Qu.:	:1.000	1st Qu.:	:1.000
Median :	:2.000	Median :	:3.000	Median :	:3.000	Median :	:2.000	Median :	:2.000
Mean :	:2.067	Mean :	:3.178	Mean :	:2.356	Mean :	:1.622	Mean :	:3.556
3rd Qu.:	:3.000	3rd Qu.:	:4.000	3rd Qu.:	:3.000	3rd Qu.:	:2.000	3rd Qu.:	:3.000
Max. :	:4.000	Max. :	:4.000	Max. :	:4.000	Max. :	:3.000	Max. :	:3.000
		I11		I12		I13		I14	
Min.	:1.000	Min.	:1.000	Min.	:1.0	Min.	:2.000	Min.	:1.000
1st Qu.:	:1.000	1st Qu.:	:2.000	1st Qu.:	:2.0	1st Qu.:	:3.000	1st Qu.:	:3.000
Median :	:2.000	Median :	:3.000	Median :	:3.0	Median :	:4.000	Median :	:3.000
Mean :	:2.022	Mean :	:2.444	Mean :	:2.4	Mean :	:3.444	Mean :	:3.556
3rd Qu.:	:2.000	3rd Qu.:	:3.000	3rd Qu.:	:3.0	3rd Qu.:	:4.000	3rd Qu.:	:5.000
Max. :	:4.000	Max. :	:4.000	Max. :	:4.0	Max. :	:4.000	Max. :	:6.000
		ITotal		I8		I9		I28	
Min.	:29.00	Min.	:1.000	Min.	:1.000	Min.	:2.000	Min.	:1.000
1st Qu.:	:34.00	1st Qu.:	:2.000	1st Qu.:	:1.000	1st Qu.:	:1.000	1st Qu.:	:1.000
Median :	:37.00	Median :	:2.000	Median :	:2.000	Median :	:3.000	Median :	:3.000
Mean :	:36.51	Mean :	:2.022	Mean :	:2.333	Mean :	:3.444	Mean :	:3.556
3rd Qu.:	:40.00	3rd Qu.:	:5.000	3rd Qu.:	:3.000	3rd Qu.:	:6.000	3rd Qu.:	:6.000
Max. :	:44.00	Max. :	:6.000	Max. :	:6.000	Max. :	:6.000	Max. :	:6.000
		I21		I22		I23		I24	
Min.	:1.000	Min.	:1.000	Min.	:1.000	Min.	:1.000	Min.	:1.000
1st Qu.:	:1.000	1st Qu.:	:1.000	1st Qu.:	:1.000	1st Qu.:	:1.000	1st Qu.:	:1.000
Median :	:1.000	Median :	:2.000	Median :	:1.000	Median :	:2.000	Median :	:2.000
Mean :	:2.067	Mean :	:2.267	Mean :	:1.667	Mean :	:2.667	Mean :	:3.000
3rd Qu.:	:3.000	3rd Qu.:	:3.000	3rd Qu.:	:2.000	3rd Qu.:	:3.000	3rd Qu.:	:3.000
Max. :	:6.000	Max. :	:7.000	Max. :	:5.000	Max. :	:7.000	Max. :	:7.000
		I25		I26		RTotal		Total	
Min.	:1.000	Min.	:1.000	Min.	:9.00	Min.	:40.00	Min.	:40.00
1st Qu.:	:1.000	1st Qu.:	:1.000	1st Qu.:	:15.00	1st Qu.:	:50.00	1st Qu.:	:50.00
Median :	:1.000	Median :	:2.000	Median :	:21.00	Median :	:59.00	Median :	:59.00
Mean :	:1.778	Mean :	:2.956	Mean :	:21.90	Mean :	:58.40	Mean :	:58.40
3rd Qu.:	:2.000	3rd Qu.:	:4.000	3rd Qu.:	:28.00	3rd Qu.:	:64.00	3rd Qu.:	:64.00
Max. :	:7.000	Max. :	:6.000	Max. :	:40.00	Max. :	:81.00	Max. :	:81.00

This is a summary of the data by column and questions.

```
ggplot(FaithandInsight, aes(x = Subnum, y = Total)) +
  geom_col() +
  geom rug() +
  labs(title = "Overall Participant Scores", x = "Participant") +
  stat_smooth(method = "lm", col = "red")
```

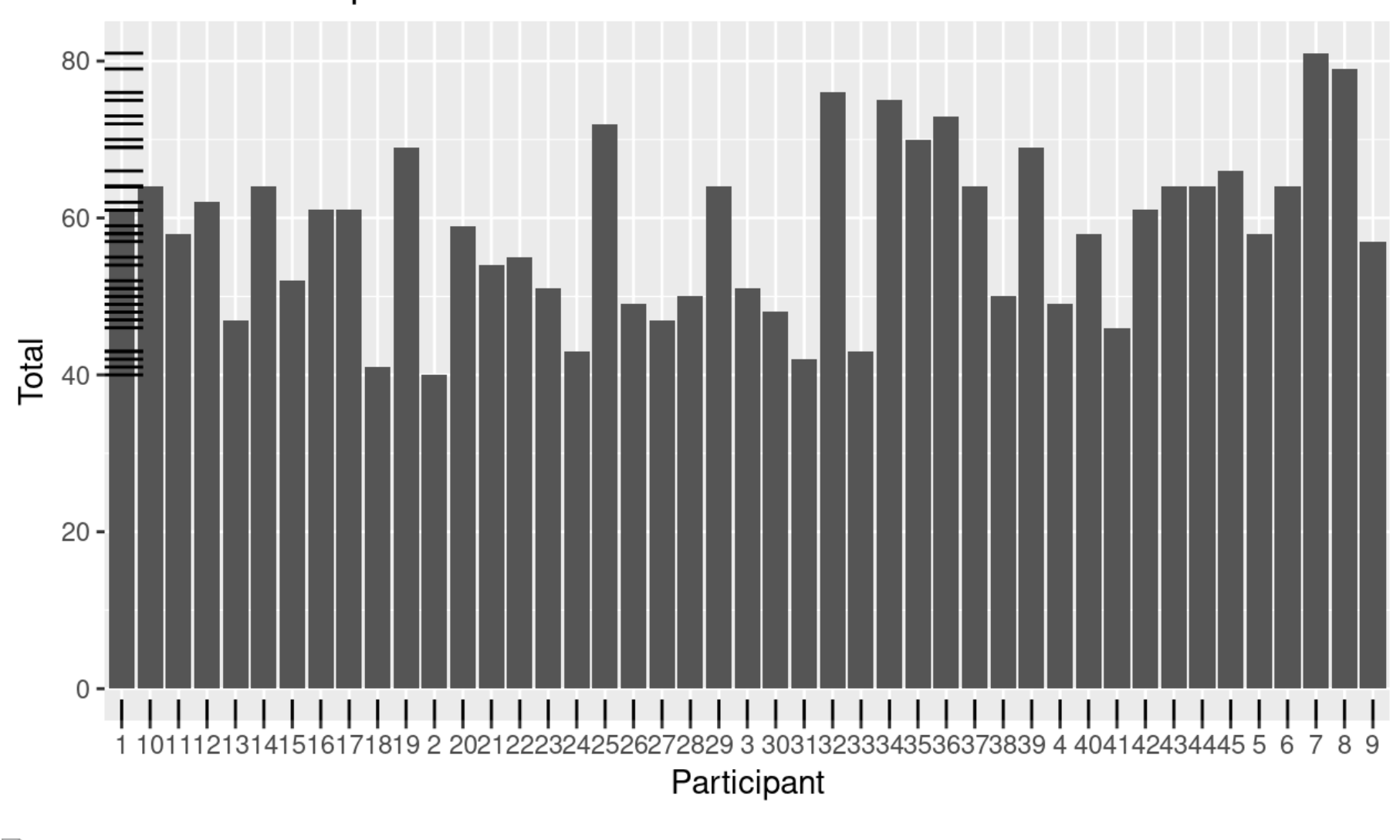


Figure 1.1 Indicates the total of each participant. The lowest scores were around 40. The highest were around 80.

```
regression.model <- lm(ITotal~ RTotal, data = FaithandInsight)
summary(lm( regression.model ))

Call:
lm(formula = ITotal ~ RTotal, data = FaithandInsight)

Residuals:
    Min       1Q   Median       3Q      Max
-7.6896 -2.6066  0.3994  3.2391  7.3816

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 36.11962    1.46466   24.661   <2e-16 ***
RTotal       0.01781    0.06103   0.292   0.772
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.947 on 43 degrees of freedom
Multiple R-squared:  0.001977, Adjusted R-squared:  -0.02123
F-statistic: 0.00519 on 1 and 43 DF,  p-value: 0.7718
```

The regression F statistic (0.00519) indicates that there is low variability. However the P value (0.7718) indicates there is not any significance of the Religiosity scores(R Total) effecting the Insight scores(I Total).

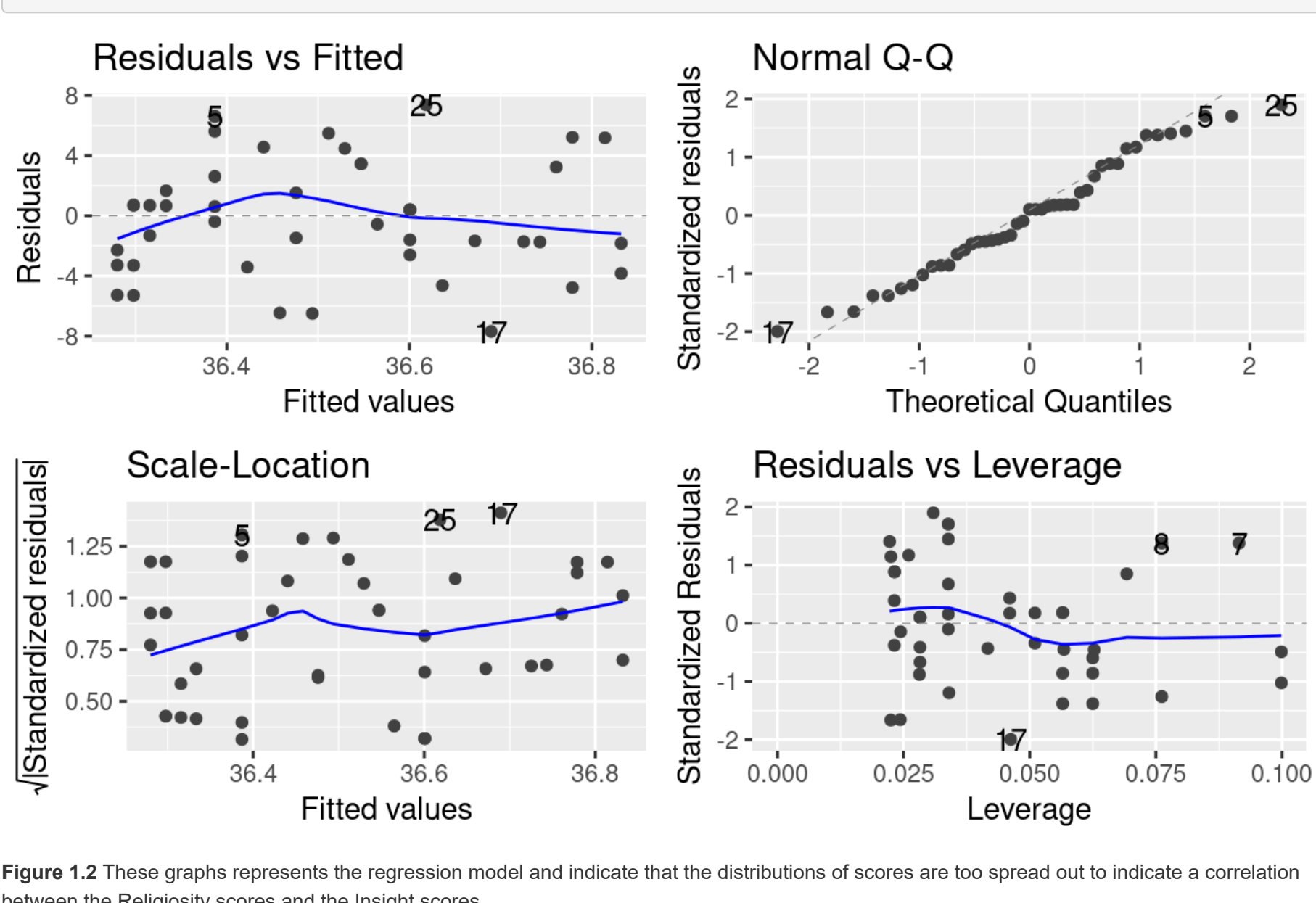


Figure 1.2 These graphs represents the regression model and indicate that the distributions of scores are too spread out to indicate a correlation between the Religiosity scores and the Insight scores.

```
aov.model <- aov(formula = ITotal ~ RTotal, data = FaithandInsight)
summary(aov.model)

              Df Sum Sq Mean Sq F value Pr(>F)
RTotal         1  1.3    1.327   0.085 0.772
Residuals     43 669.9   15.579
```

This One-Way ANOVA and the P value (0.772) indicate there is not any significance of the Religiosity scores (R Total) effecting the Insight scores (I Total).

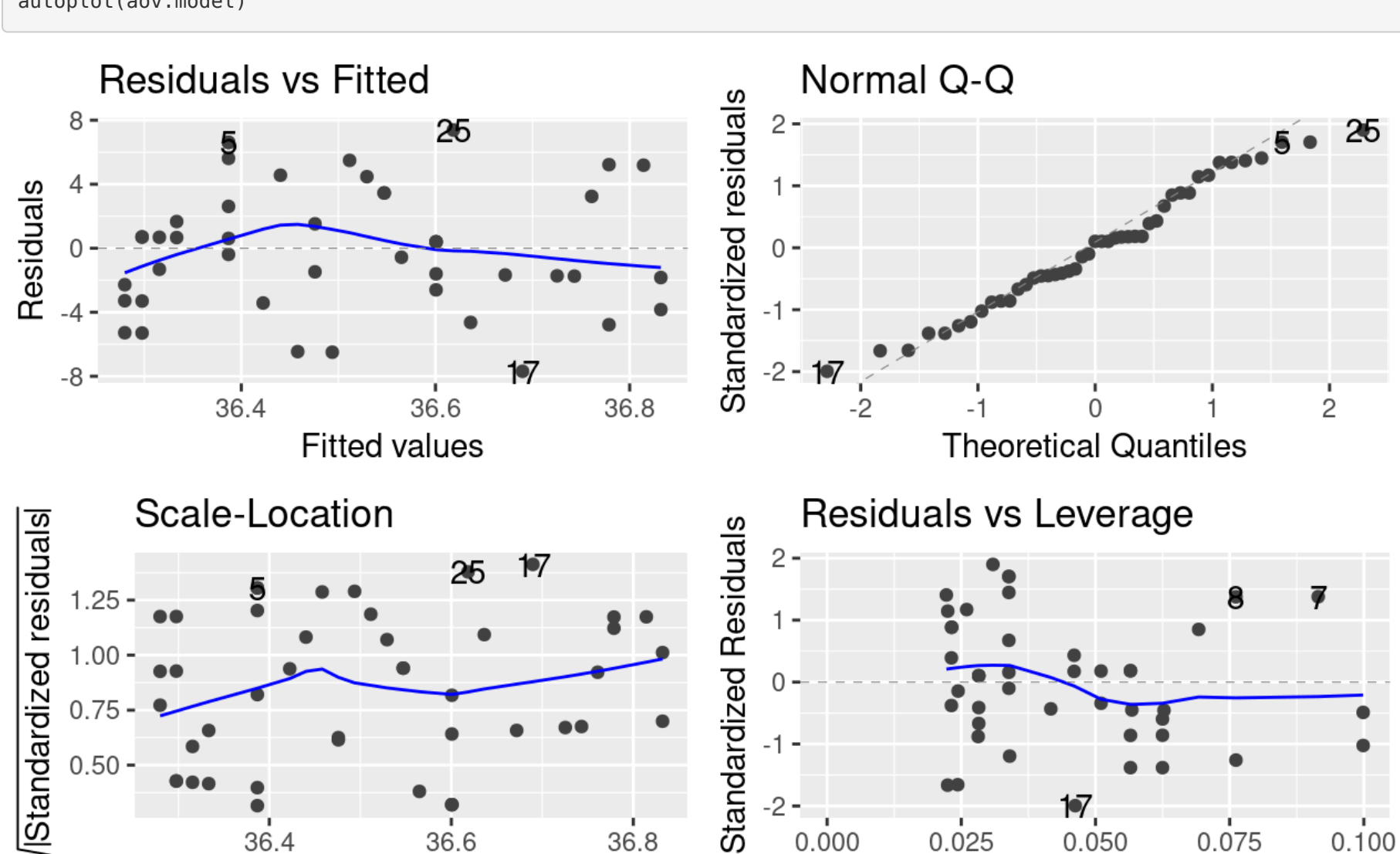


Figure 1.3 These graphs represent the One-Way ANOVA and indicate that the distributions of scores are too spread out to indicate a correlation between the Religiosity scores and the Insight scores.

```
aov.model2 <- aov(formula = ITotal ~ RTotal + Age + Gender, data = FaithandInsight)
summary.aov(aov.model2)

              Df Sum Sq Mean Sq F value Pr(>F)
RTotal         1  1.3    1.327   0.085 0.773
Age             1  8.0    7.971   0.508 0.480
Gender          1 18.2   18.210   1.169 0.288
Residuals     41 643.7   15.701
```

The Three-way ANOVA and the Religiosity (R Total) P value of (.773), Age P value of (.480), and Gender P value of (.288) indicate there is not any significance of the Religiosity, Age, or Gender Variables effecting the Insight scores(I Total).

```
ggplot(aov.model2, aes(group = Age, x = RTotal, y = ITotal, color = factor(Age, labels = c("18-30",
"31-40", "41-50", "51-60", "60+")))) +
  theme(legend.text = element_text(colour="black", size = 16, face = "bold")) +
  theme(legend.background = element_rect(colour = 'dark blue', fill = 'green', size = 3)) +
  geom_boxplot() +
  labs(color = "Age") +
  labs(title = "Scores by Age Groups")
```

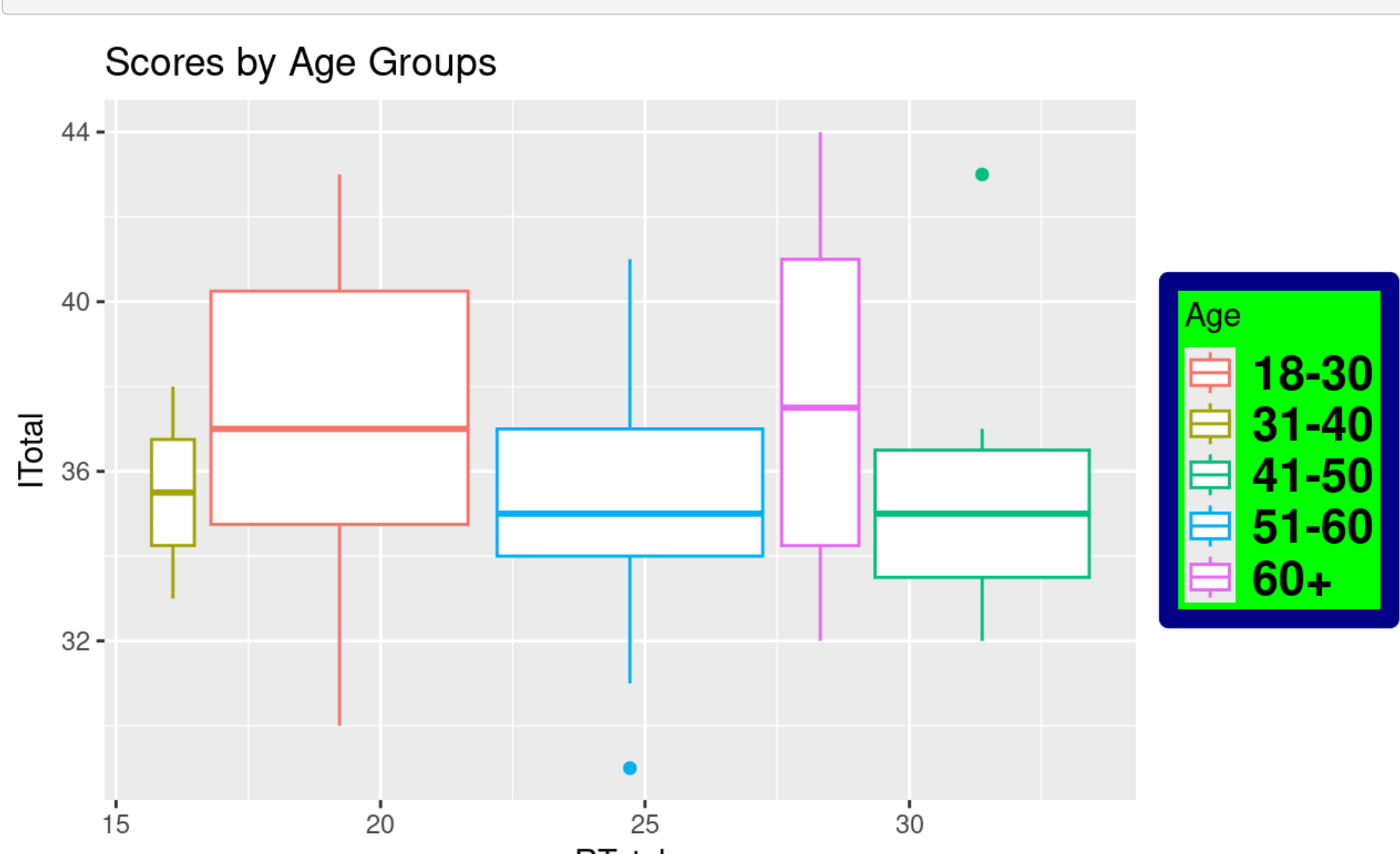


Figure 1.4 Represents the Three-Way ANOVA and indicates that the highest scoring group of participants were in the 18-30 range and the least in the 31-40 range. Those in the 18-30 range scored higher on Insight. Those in the 18-30 were second highest on Insight. There is little variation from the mean.

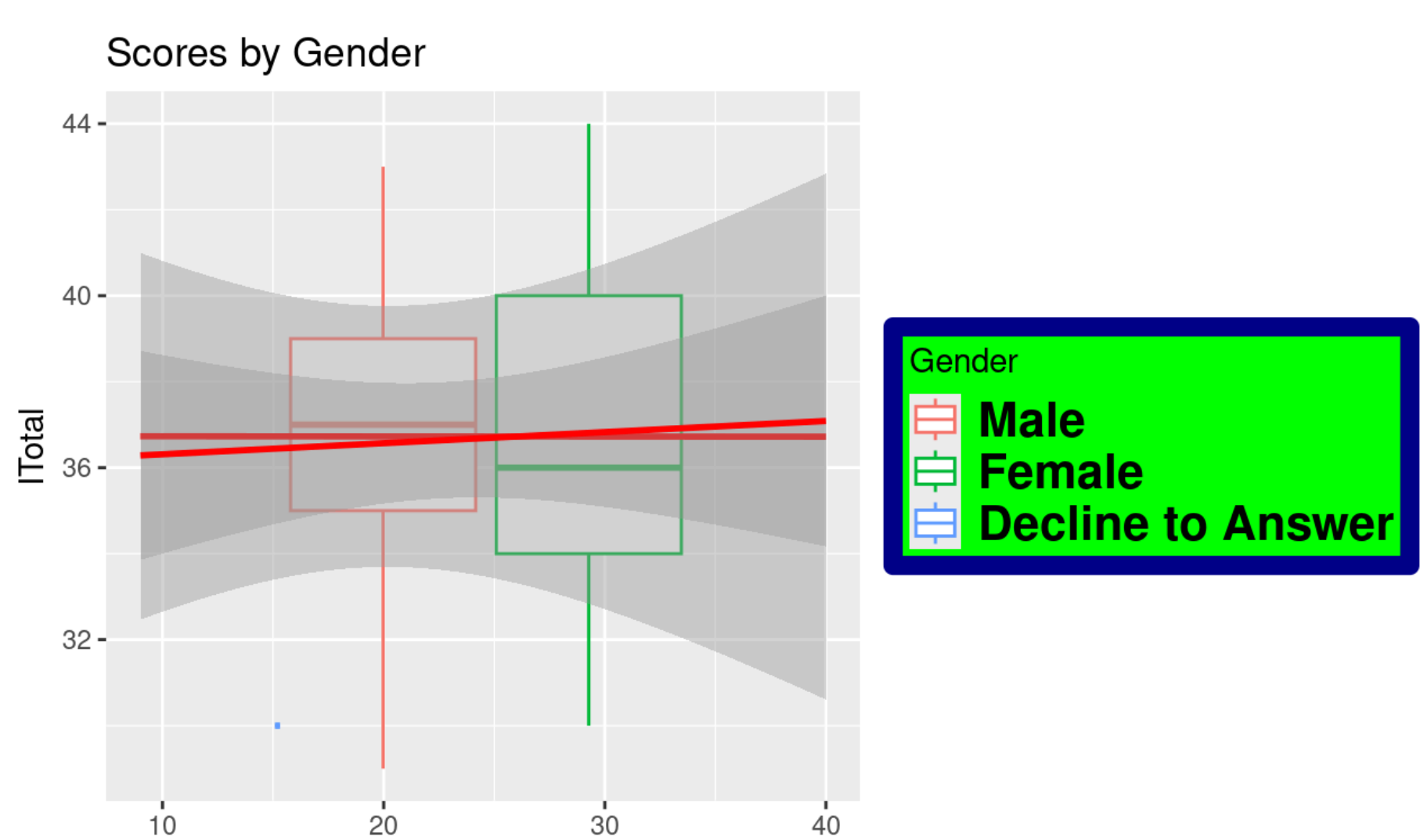


Figure 1.5 Represents the Three-Way ANOVA with a focus on Gender and indicates that female participants scored higher on Insight. There is little variation from the mean.

```
manova.model <- manova(cbind(ITotal, RTotal) ~ Age*Gender, data = FaithandInsight)
summary(manova.model, test = "Wilks")

              Df  Wilks approx F num Df den Df Pr(>F)
Age             1  0.9400    1.1864      2    40 0.3158
Gender          1  0.94253  1.2195      2    40 0.3061
Age:Gender      1  0.94715  1.1159      2    40 0.3376
Residuals      41
```

This Two-way MANOVA indicates that the variables of Age and Gender fit the model well with little discrimination as indicated by the Wilks test scores being under 1. However the P value for Gender (.3061), and the P value for Age (.3158) indicate there is no significant in any of the variables to Religiosity or Insight.

```
ggplot(manova.model, aes(x = Age, y = Gender, color = factor(Gender, labels = c("Male",
"Female", "Decline to Answer")))) +
  geom_violin() +
  scale_x_continuous(labels = c("1" = "18-30", "2" = "31-40", "3" = "41-50", "4" = "51-60", "5" = "60+")) +
  labs(color = "Gender") +
  labs(title = "Gender Focus") +
  stat_smooth(method = "aov", col = "red")
```

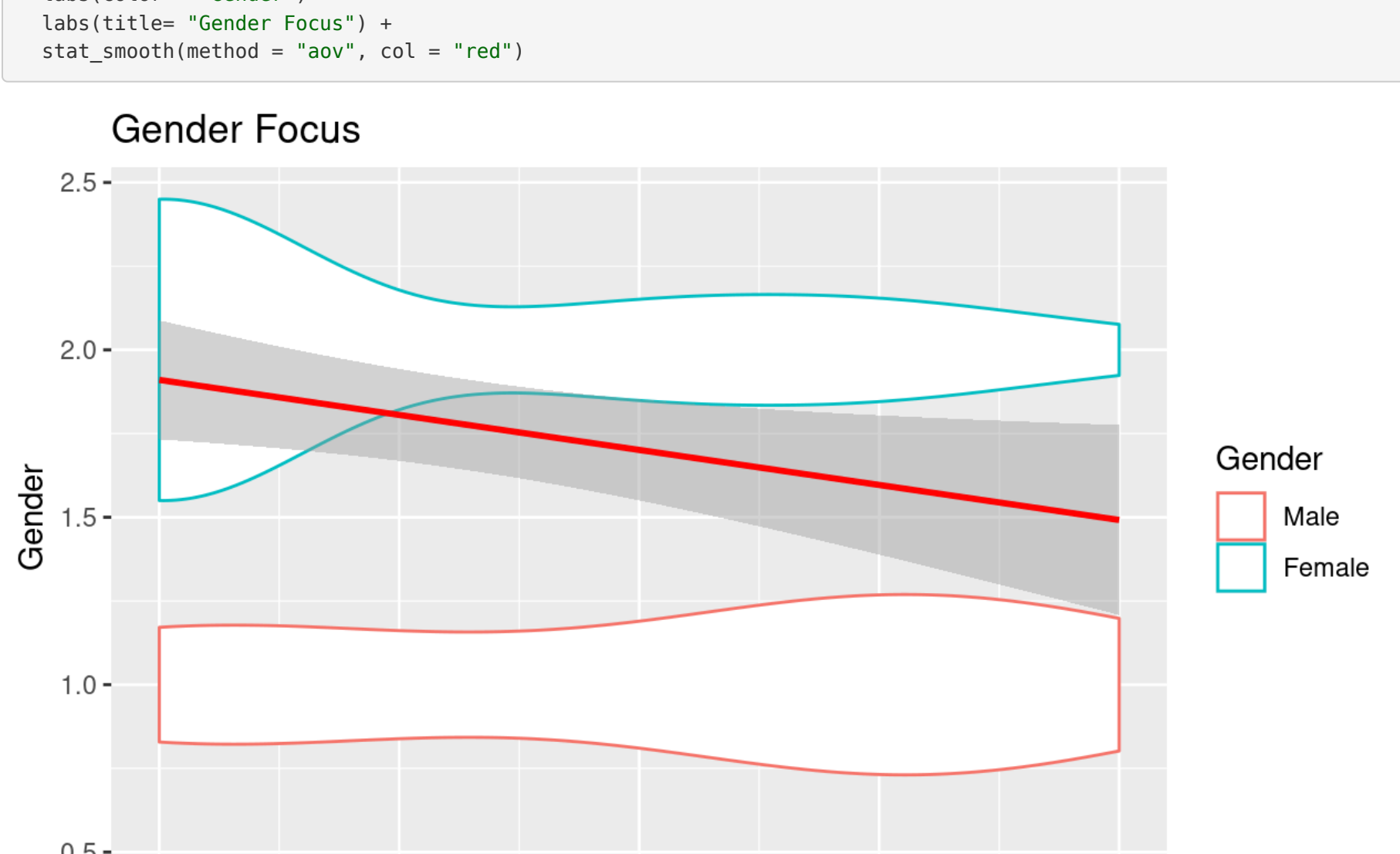


Figure 1.6 Represents the MANOVA measuring the relationship of Age and Gender to Religiosity and Insight. It appears that females in the 18-31 range scored the highest on Religiosity and Insight. Males from the 41-60 range scored highest for their group.

```
ggplot(manova.model, aes(x = Age, y = Gender, color = factor(Age, labels = c("18-30", "31-40", "41-50", "51-60",
"60+")))) +
  geom_violin() +
  labs(color = "Age") +
  labs(title = "Age Focus") +
  stat_smooth(method = "aov", col = "red")
```

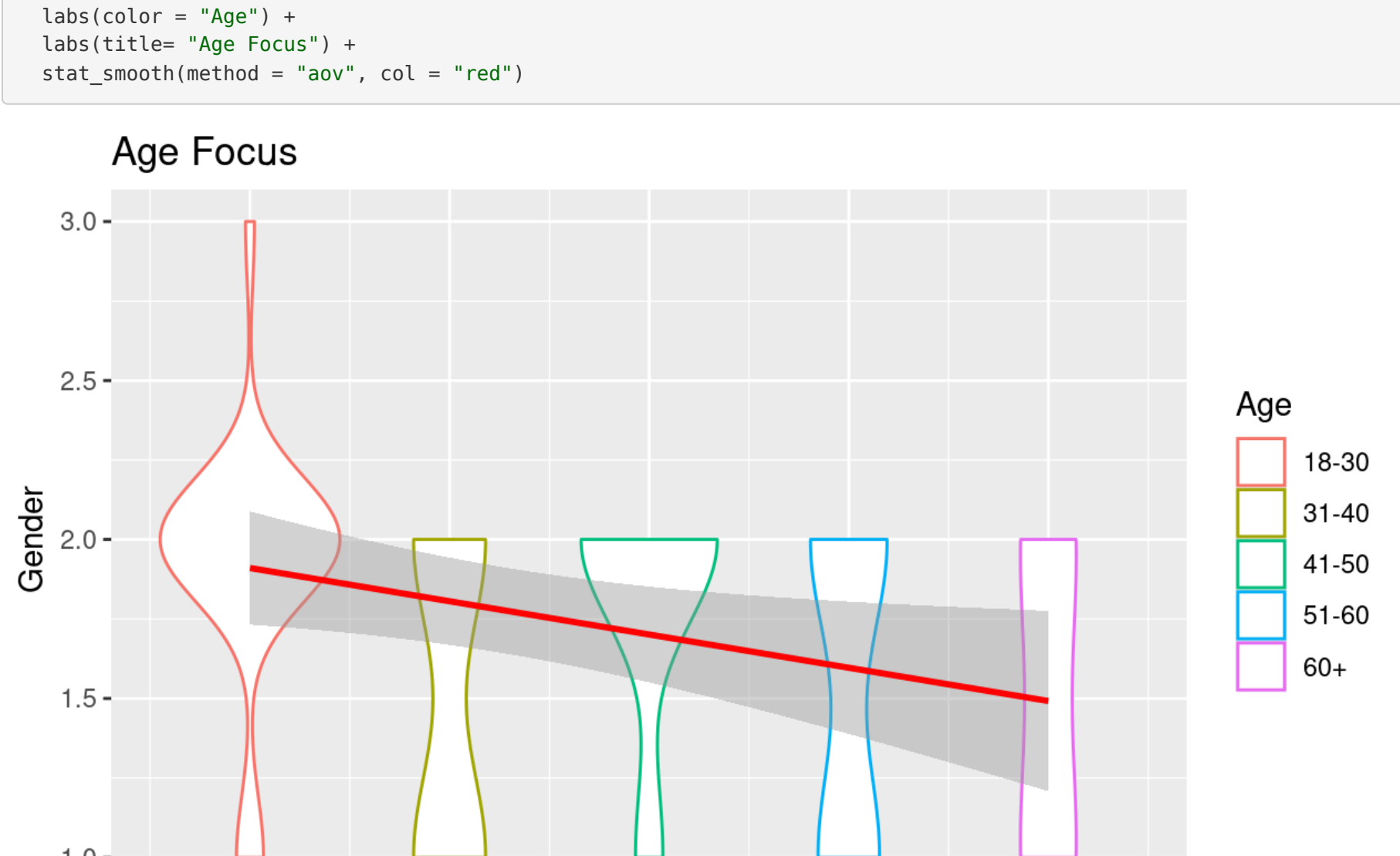


Figure 1.7 Represents the Age perspective of the above MANOVA and indicates that those in the 41-50 range scored the lowest on Religiosity and Insight while those in the 18-30 range scored the highest.

Discussion

The Overall goal of this study is to see if there is a relationship between ones Insight and how Religious the participants are. In this Regard there was no evidence found to support that hypothesis. With the small sample studied it was discovered that participants in the 18-30 age group scored the highest on Insight and Religiosity. The 41-50 and the group that scored the least on Insight and Religiosity were those in the age group of 31-40. It was also found that females scored higher on both Religiosity and Insight scales. There was no significant correlation found between Age or Gender on Religiosity or Insight. In the future this study could yield better results with a larger sample size and a more diverse distribution among age. In the future switching the primary scale looked at from the Insight scores to the Religiosity scores would be beneficial for a better overall picture.

Work Cited